Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Backend software engineer pursuing an MSCS with 2+ years building cloud-backed APIs, reliable data workflows, automated tests, and real-time operational systems across Python, JavaScript, and distributed services.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Azure SQL-backed backend services** for 500+ users, translating product requirements into APIs, validation logic, schema updates, and production releases across cloud-hosted data workflows and internal tools.
- Improved **service reliability and performance** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** while supporting operational debugging and handoffs.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **cross-platform integration services** using Python, SQL, AWS Glue, and S3, improving response time by **30%** while supporting distributed APIs, data movement, and reporting workflows reliably.
- Maintained **20+ automated validation suites** with Jenkins, PyTest, and Mockito, increasing CI reliability by **68%** and reducing regressions across Windows, Linux, and macOS environments consistently for releases.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **Python services and REST APIs** handling 10,000+ daily requests, integrating uploads, OCR processing, persistence workflows, and automated checks for production reliability under active usage daily.
- Debugged **runtime and workflow defects** across upload, OCR, persistence, and grading failures, documenting fixes and reducing feedback turnaround by **60%** for production users and operators consistently.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **evaluation pipelines** across 100,000+ records, producing structured metrics, logs, and reports for reproducible analysis, anomaly investigation, and reliable model validation workflows at scale for researchers.
- Documented **technical workflows and review outputs** for researchers, turning ad hoc experiments into repeatable processes with clearer debugging notes, validation artifacts, and handoffs across teams during reviews.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **REST and WebSocket modules** over SQL/NoSQL persistence, supporting 2M+ synchronized records across integrations, dashboards, and operational workflows for high-volume users and systems during daily internal operations.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting, and team handoffs during releases and validations across environments.

Skills

Languages: **Python**, JavaScript, TypeScript, SQL, Java, C++, Bash, HTML/CSS
Backend: REST APIs, Flask, FastAPI, Node.js, WebSockets, object-oriented design, distributed workflows
Cloud/Data: Azure SQL, AWS Glue, S3, PostgreSQL, MySQL, MongoDB, ETL, background jobs, structured logging
Testing: PyTest, Mockito, Jenkins, CI/CD, automated validation, debugging, code reviews, documentation
Practices: Git/GitHub, Agile/Scrum, AI-assisted development, operational reliability, incremental delivery

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built a **secure real-time platform** supporting 50K+ users and 1M+ transactions, integrating live updates, tokenized access controls, CI/CD workflows, and operational visibility dashboards for teams and operators.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed a **containerized REST processing API** with AWS deployment, CI/CD releases, validation checks, and reproducible execution, achieving **87%** classification accuracy with diagnostic reporting and operational logs.

CPU Scheduling Simulator Live | *C++, React, JavaScript, OS Algorithms*
- Implemented **FCFS, SJF, SRT, RR, priority, and MLQ** schedulers with C++ logic, debugging workflows, validation, timing analysis, and interactive visualization across simulated workloads reliably for testing exercises.